

8994
6371
12705

11719
14766
7028
4580
8280
13928
13368
12612
11467
10550

case targeting 22 short fragments encompassing haplogroup-specific SNPs in the coding region (as defined in Behai' et al., 2007; Figure 3). A SBE reaction follows, where oligonucleotides designed to hybridise one base pair upstream of each SNP of interest are used as primers for the addition of a single dideoxyribonucleoside triphosphate (ddNTP). Each ddNTP is labelled with a different fluorochrom to allow detection of the SNPs after fragment length-based electrophoresis separation of the 22 amplicons.

The GenoCoRe22 reaction is particularly relevant to the study of ancient mtDNA for several reasons. First, because the GenoCoRe22 targets 22 SNPs in a single reaction it is a time and cost-efficient method that uses very small amounts of valuable aDNA extracts. Because it limits the number of PCRs necessary to type 22 SNPs, GenoCoRe22 also reduces the risk of introducing modern contamination in the typing reactions (as opposed to 22 single reactions). As the multiplex is designed to target short DNA fragments spanning sizes from 60 bp to 80 bp, it is suitable for highly degraded DNA. Finally, the GenoCoRe22 reaction is used here as an internal control. The concordance between the haplogroup assessments independently obtained from the analysis of HVR-I sequences and SNaPshot profiles provides a key means to authenticate the aDNA results. In addition, because of the hierarchical nature of its design, i.e., typing of SNPs diagnostic of mtDNA macro-haplogroups, haplogroups and sub-haplogroups, the GenoCore22 reaction allows the detection of contamination and/or DNA damage through the observation of superimposed and/or phylogenetically inconsistent profiles.

-L0
'1
-H2
6
"L3'
4
-M
"C
-D
-A
"N1
-W
-X
■R
■R
O
■HV
■H
-V
■B
"R9

Figure 3: Schematic representation of the human mitochondrial phylogeny indicating the positions of the Single Nucleotide Polymorphisms typed in the GenoCore22 reaction.